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الجامعة الإسلامية العالمية شيتاغونغ
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PROJECT REPORT
On
PASSWORD MANAGER

Course Title: Tools and Technologies for Internet Programming

Course Code: CSE-3532

Submitted To:

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Introduction:

With the rise in the number of online accounts, remembering passwords has become difficult. "PassOP" is a secure and easy-to-use password manager that helps users store and manage their credentials for different websites. Built using the **MERN stack** (MongoDB, Express.js, React.js, Node.js) and styled with **Tailwind CSS**, this project focuses on offering a minimal, secure, and user-friendly interface.

Objective:

- To develop a password manager with features like storing, updating, viewing, and deleting passwords.
- To ensure security and authentication using encryption and tokens.
- To provide a responsive and intuitive UI using React and Tailwind.

Technologies Used:

Technology	Purpose
React.js	Frontend UI
Tailwind CSS	Styling and responsive layout
Express.js	Backend API services
MongoDB	NoSQL database for storing user credentials
Node.js	Server environment

Why It Is Useful:

1. Centralized Password Storage

Users often have dozens (or hundreds) of online accounts. Your app allows them to:

- Store all their credentials securely in one place.
- Avoid forgetting passwords or reusing weak ones

2. Enhances Security Awareness

With features like:

- **Security audit** (weak, reused, strong password metrics),
- **Security score**, and
- **Recommendations** (e.g., "Strengthen Now", "Enable 2FA"),

...it educates users about password hygiene and encourages safer habits.

3. Password Generator

The built-in generator helps users:

- Create strong, unique passwords.
- Customize them with length, symbols, numbers, etc.
- Copy them easily with a single click.

4. Secure Notes Storage

Users can:

- Store sensitive notes (like PINs, license keys, recovery codes),
- Keep them encrypted and private,
- Access them across sessions if saved to backend.

⚙️ 5. Settings for Customization

Your Settings.jsx allows users to:

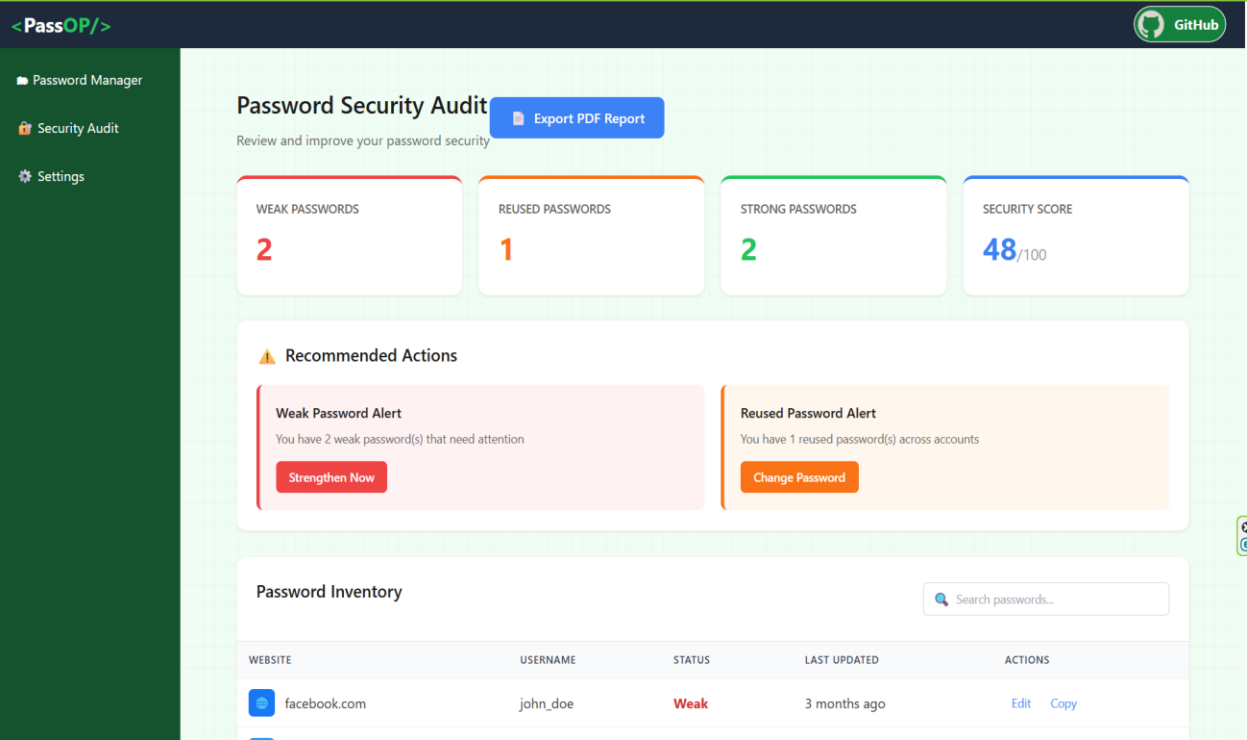
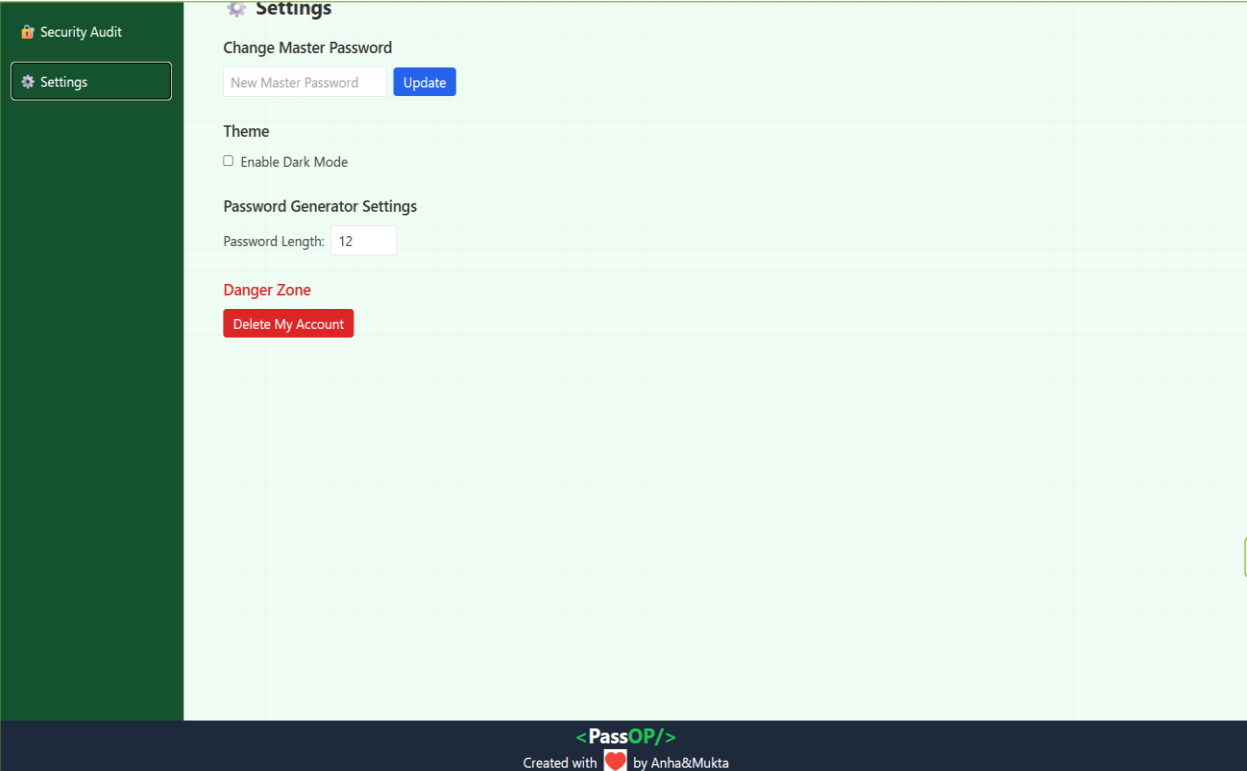
- Change master password,
- Toggle dark mode for accessibility,
- Set generator defaults,
- Delete account if needed.

🌐 6. Syncs with Backend (MongoDB + Express)

- Notes and passwords persist across sessions.
- Data is stored securely on the backend.
- Avoids localStorage limitations.

The screenshot displays the PassOP web application interface. On the left is a dark green sidebar with navigation links: Password Manager, Security Audit, Settings, and Secure Notes. The main content area has a light green background with a grid pattern. At the top, it shows the PassOP logo and a GitHub link. Below this is a form to add a new password with fields for 'Enter website URL', 'Enter Username', and 'Enter Password', followed by a 'Save Password' button. A section titled 'Your Passwords' contains a table with four entries.

SITE	USERNAME	PASSWORD	ACTIONS
www.youtube.com	ummehabibaanha6688@gmail.com	*****	🔍 🗑️
www.instagram.com	meharun mukta	*****	🔍 🗑️
www.linkedin.com	ummehabibaanha	*****	🔍 🗑️
www.github.com	ummehabibaanha6688@gmail.com	****	🔍 🗑️



facebook.com

john_doe

Weak

3 months ago

Edit

Copy

twitter.com

john_doe

Reused

2 weeks ago

Edit

Copy

github.com

john123

Strong

1 week ago

Edit

Copy

netflix.com

john_doe

Strong

1 month ago

Edit

Copy

oldforum.com

john_doe

Weak

5 months ago

Edit

Copy

Security Recommendations

Strengthen Weak Passwords

You have 2 weak passwords that need to be strengthened.

Fix Now

Replace Reused Passwords

You have 1 reused passwords across multiple accounts.

Fix Now

Enable Two-Factor Authentication

Add an extra layer of security to your important accounts.

Learn More

Password Generator

g3N3r@T3dP@ssw0rD

g3N3r@T3dP@ssw0rD

Password Length:

16

☒ Include Uppercase

☒ Include Lowercase

☒ Include Numbers

☒ Include Symbols

Generate New Password

<PassOP/>

GitHub

Password Manager

Security Audit

Settings

Secure Notes

Secure Notes

Note Title

Note Content

Save Note

Climate Change

The long-term shift in global or regional climate patterns, largely driven by human activities like fossil fuel burning, resulting in rising global temperatures and extreme weather events.

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Cybersecurity

The practice of protecting systems, networks, and programs from digital attacks, which can result in data breaches, identity theft, or financial loss.

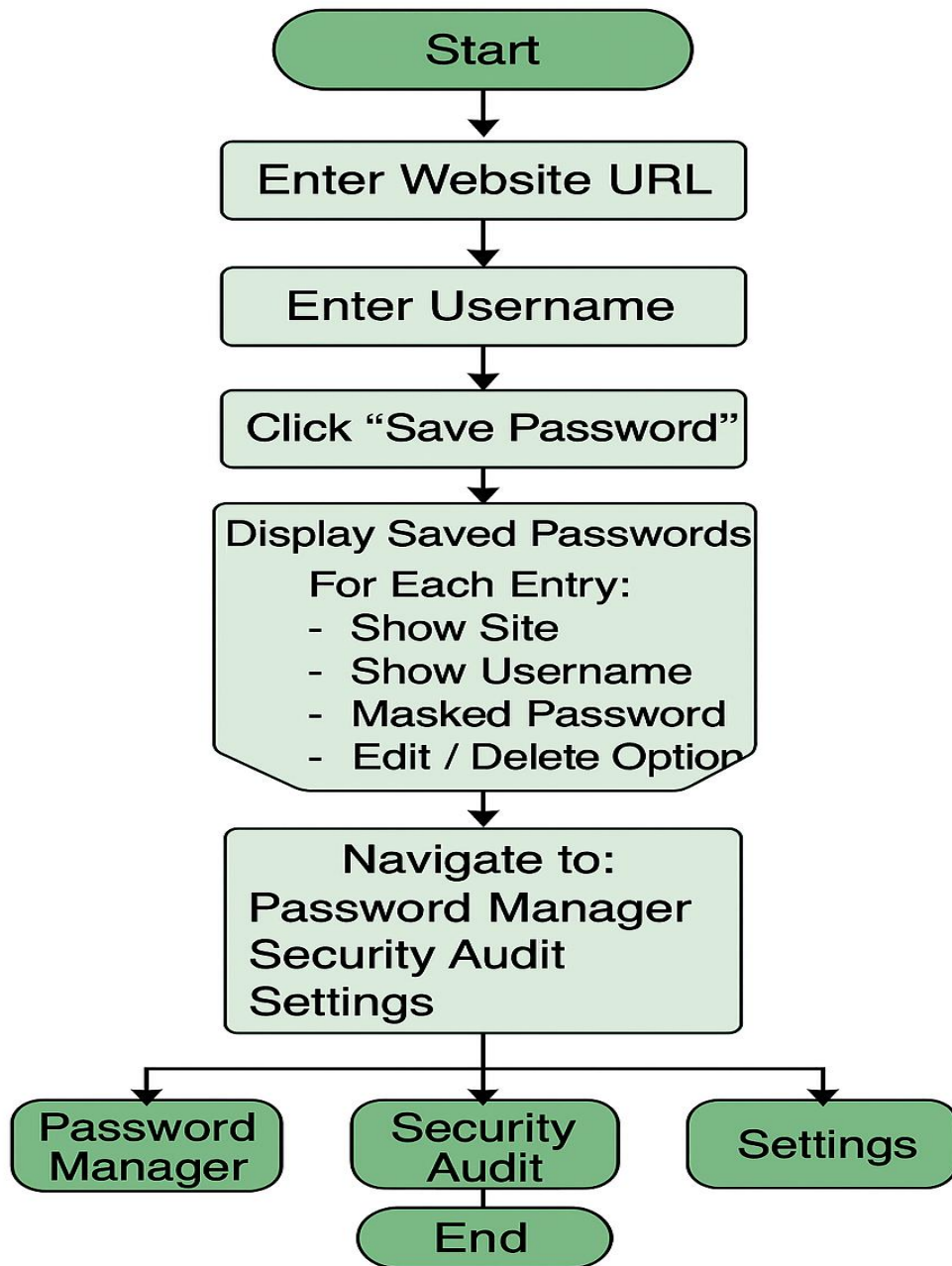
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Artificial Intelligence (AI)

AI involves machines performing tasks that typically require human intelligence, such as learning, problem-solving, and decision-making.

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Flowchart



Team Contributions:

Umme Habiba Anha (ID: C231441):

Implemented Edit and Delete Password functionalities.

Worked on parts of the Home Page layout and logic.

Developed the Note Page, allowing users to:

Store sensitive notes (like PINs, license keys, recovery codes)

Keep notes encrypted and private

Access notes across sessions when saved to the backend (MongoDB)

Handled saving passwords to Local Storage for quick client-side access.

Integrated MongoDB backend to securely save user passwords.

Meharunnasa Mukta (ID: C231446):

Developed the Security Audit Page, featuring:

Weak password alerts

Reused password detection

Secure storage indicators

Built the Settings Page, including:

Master password update

Dark mode toggle

Password strength configuration

Delete account option

Contributed to parts of the Home Page design and logic.